BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ABBOTTABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

BIOLOGY

Past Paper 2023 • Paper Code: 11231

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** 11231

Note: Attempt all questions of Section-A by filling the corresponding bubble on the MCQs RESPONSE SHEET. It is mandatory to return the attempted MCQs sheet to the Superintendent within given time.

Q1. Amino acid mainly differ from each other by the difference in their .....

(A) R-group
(B) amino group
(C) carboxyl group
(D) alpha group

Q2. DNA is more or less present in all cells except .....

(A) nucleus
(B) mitochondria
(C) chloroplast
(D) ribosomes

Q3. Identify the following which is not a limiting factor for photosynthesis:

(A) light intensity
(B) temperature
(C) concentration of carbon dioxide
(D) concentration of oxygen

Q4. HIV viruses are ..... containing.

(A) RNA
(B) DNA
(C) gene
(D) protein

Q5. Bacterium capsules are formed from .....

(A) polysaccharides and protein
(B) polysaccharides and genes
(C) 2-units of polysaccharides
(D) none of these

Q6. Live spores have been recorded from the intestine of .....

(A) human
(B) rhynia
(C) mummies
(D) psilotum

Q7. The main causes of pneumonia are .....

(A) mycobacterium
(B) xanthomonas
(C) salmonella
(D) streptococcus

Q8. Agrobacterium tumifaciens causes .....

(A) bacterial gall
(B) late blight
(C) bacterial blight
(D) red rot

Q9. Biological name of yeast is .....

(A) claviceps
(B) pencillium
(C) saccharomyces
(D) cephalorus

Q10. Fungi and algae are evolved in the era of .....

(A) coenozoic
(B) mezozoic
(C) paleozoic
(D) proterozoic

Q11. Earthworm belongs to phylum .....

(A) annelida
(B) mollusca
(C) nematoda
(D) porifera

Q12. ..... is typical supporting tissues of the primary plant body and growing plant parts.

(A) parenchyma
(B) collenchyma
(C) sclerenchyma
(D) all of these

Q13. Calamites is the extinct group of .....

(A) psilopsida
(B) lycopsida
(C) sphenopsida
(D) pteropsida

Q14. Association of lichen and fungi is the example of .....

(A) mutualism
(B) parasitism
(C) predation
(D) commensalism

Q15. Translocation in bacteria was discovered by .....

(A) Lederberge & Zinder
(B) Lederberge & Tatum
(C) Harshay & Chase
(D) all of these

Q16. Heart is derived from ..... word kardia:

(A) Greek
(B) Latin
(C) English
(D) Anatolian

Q17. The primary growth of a plant is due to the action of .....

(A) lateral meristem
(B) apical meristem
(C) cork cambium
(D) vascular cambium

Q18. A bacterium with a cluster of flagella is called .....

(A) peritrichous
(B) amphitrichous
(C) monotrichous
(D) lophotrichous

SECTION-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 40 | **Paper Code:** 11231

Q1. Attempt any ten of the following short questions. Each question carries 4 marks.

(i) Give a brief description of carbohydrates.

(ii) Highlight the types of inhibitors in enzyme.

(iii) Describe chlorophyll in a few lines.

(iv) Briefly discuss "Cyclic Electron Transport".

(v) What do you know about HIV AIDS?

(vi) Briefly explain the structure of bacteria.

(vii) What is edible fungi? Discuss briefly.

(viii) Give a brief account of psilopsida.

(ix) Briefly explain phylum arthropoda.

(x) Discuss evolution in echinoderms.

(xi) Briefly describe starch-sugar theory.

(xii) What do you know about ethylene?

(xiii) Explain interferon in a few lines.

SECTION-C

Maximum Marks: 27

Note: Attempt any three of the following questions. Each question carries 9 marks.

Q2.

(i) Describe the structure of bacteriophage virus.

(ii) Explain osmoregulation in plants.

Q3. Define immunity. Explain in detail the different types of immunity.

Q4.

(i) Define and explain Kreb's Cycle.

(ii) Write down the importance of gymnosperm.

Q5. Describe any two of the following topics:

(i) Function of Heart

(ii) Food Poisoning

(iii) Vernalization